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Programming for Data Analytics 2

Part 2 – Classification and Model Improvement

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Introduction

Machine learning is increasingly used to support data-driven decision-making across various industries, including healthcare. One important category of machine learning is classification, where the goal is to assign observations to predefined categories based on their features.

This study focuses on a binary classification problem in the medical domain, where tumor samples are classified as either malignant or benign. Classification techniques are appropriate because the target variable consists of discrete categories rather than continuous values.

The aim of this study is to develop a predictive model that can accurately classify tumor samples based on their characteristics. To achieve this, a structured approach is followed, including dataset selection, exploratory data analysis (EDA), feature selection, model training, and evaluation.

1. Dataset Selection and Justification

The dataset selected for this study is the **Breast Cancer Wisconsin (Diagnostic) dataset**, obtained from **Kaggle**, originally sourced from the **UCI Machine Learning Repository** (UCI Machine Learning Repository, 1995). This dataset was chosen due to its suitability for classification modelling and its relevance to real-world healthcare applications.

The dataset contains **569 observations and 30 numerical features**, all derived from digitized images of breast mass tissue samples. These features describe characteristics of cell nuclei, such as radius, texture, perimeter, and concavity. The target variable is binary, where tumors are classified as malignant or benign, making the dataset highly suitable for classification algorithms such as Logistic Regression, Decision Trees, and Random Forests (Scikit-learn, 2024).

According to Kaggle, the dataset has a **high usability score**, indicating that it is well-structured, clearly documented, and suitable for machine learning analysis. A high usability score suggests that the dataset is easy to work with, contains minimal noise, and requires limited preprocessing.

One of the key strengths of the dataset is that it contains **no missing values**, which simplifies data preparation and allows focus on model development. In addition, the features are meaningful and derived from real medical measurements, providing a strong relationship between input variables and the classification outcome.

The dataset is also widely used as a **benchmark dataset** in machine learning research and education (GeeksforGeeks, 2025). This improves reliability and allows results to be compared with other studies.

Furthermore, breast cancer is a major global health concern, and early detection plays a critical role in improving patient outcomes. Therefore, using this dataset ensures that the analysis addresses a meaningful real-world problem with practical significance.

Dataset Pitfalls and Mitigation

Although the dataset is well-structured and suitable for classification, it is important to identify potential limitations and describe how they are addressed as seen in table1.

Table 1: Dataset Pitfalls and Mitigation Strategies

Potential Pitfall	Description	Mitigation Strategy
Multicollinearity	Several features (e.g. radius, perimeter, area) are highly correlated, providing similar information	Correlation analysis and feature selection will be used to remove redundant features
Class imbalance	There are more benign cases than malignant cases	Evaluation metrics such as precision, recall, and confusion matrix will be used instead of relying on accuracy alone
Feature variability	Some features have large value ranges and skewed distributions	Appropriate algorithms (Random Forest) and EDA techniques will be used to handle variation and outliers

Despite these limitations, the dataset remains **highly suitable for classification modelling** due to its quality, structure, and relevance.

2. Algorithm Selection and Justification

Several classification algorithms were considered for this study, including Logistic Regression, K-Nearest Neighbours (KNN), Decision Trees, and Random Forest. However, Random Forest was selected as the most suitable algorithm due to its robustness, ability to handle complex data, and strong predictive performance.

Logistic Regression assumes a linear relationship between the independent variables and the log-odds of the dependent variable (James et al., 2021). While this makes it simple and

interpretable, it may limit its ability to capture more complex, non-linear patterns present in the Breast Cancer Wisconsin dataset. Similarly, K-Nearest Neighbours (KNN) is sensitive to noise and requires feature scaling, which can affect its performance (Han et al., 2011). Decision Trees, although capable of modelling non-linear relationships, are prone to overfitting when used as a single model.

In comparison, Random Forest builds on the concept of Decision Trees by combining multiple trees using ensemble learning, which reduces overfitting and improves generalisation (Breiman, 2001). This makes it particularly suitable for datasets with multiple features, such as the Breast Cancer Wisconsin dataset. In addition, Random Forest can effectively handle non-linear relationships between variables, which are common in medical data.

Another advantage of Random Forest is that it requires minimal preprocessing, as it does not rely on feature scaling. It is also less sensitive to noise compared to algorithms such as KNN (Han et al., 2011). Furthermore, Random Forest provides feature importance measures, allowing for better interpretation of which variables contribute most to the classification outcome (Scikit-learn, 2024).

Despite these advantages, Random Forest has some limitations. The model can be less interpretable than simpler models such as Logistic Regression, and it may require more computational resources when a large number of trees is used. However, its ability to deliver high accuracy, handle complex feature interactions, and generalize well to unseen data makes it the most appropriate choice for this study.

3. Planning the Analysis

A structured analysis plan was developed to ensure that the dataset is properly explored, prepared, and used to train an effective classification model. The plan follows a systematic workflow consisting of exploratory data analysis (EDA), data preprocessing, feature selection, model training, evaluation, and reporting.

3.1 Exploratory Data Analysis (EDA)

The first step involves conducting exploratory data analysis to understand the structure and characteristics of the dataset. This includes examining the number of observations, features, and data types. Summary statistics will be generated to analyze the distribution, central tendency, and variation of each feature.

Visualizations such as histograms, box plots, and correlation heatmaps will be used to identify patterns, detect outliers, and explore relationships between variables. Additionally, the class distribution (malignant vs benign) will be analyzed to determine whether class imbalance exists, as this may influence model performance.

3.2 Data Preprocessing

The dataset will be prepared for modelling by separating it into input features (X) and the target variable (y). The data will then be split into training and testing sets using an 80/20 split.

A fixed random state will be used to ensure reproducibility of results and consistent model behavior across multiple runs.

3.3 Feature Selection

Feature selection will be conducted to identify the most relevant variables contributing to the classification task. This will involve:

- Correlation analysis to detect relationships between features
- Using feature importance from the Random Forest model
- Removing redundant or less informative variables

This step helps reduce multicollinearity, improve model performance, and decrease computational complexity.

3.4 Model Training

The Random Forest classifier will be trained using the training dataset. To optimize performance, hyperparameter tuning will be applied by adjusting parameters such as:

- Number of trees (*n_estimators*)
- Maximum tree depth (*max_depth*)

Techniques such as GridSearchCV may be used to identify optimal parameter values and improve the model's predictive accuracy (Scikit-learn, 2024).

3.5 Model Evaluation

The model will be evaluated using the test dataset to assess its predictive performance. Evaluation metrics will include:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Multiple evaluation metrics will be used instead of relying solely on accuracy, as this provides a more comprehensive assessment of model performance, especially in datasets with class imbalance where correctly identifying malignant cases is critical (Han *et al.*, 2011).

3.6 Model Improvement

To further enhance the model, additional techniques will be considered, including:

- Hyperparameter tuning

- Cross-validation (e.g. k-fold cross-validation)

These techniques help improve model generalization and reduce the risk of overfitting.

3.7 Reporting

The final step involves documenting the analysis process, including findings from EDA, model performance, and any improvements. Results will be presented using visualizations and clear interpretation to communicate insights effectively.

4 Conduct Analysis

4.1 Exploratory Data Analysis (EDA)

Dataset Overview

The dataset used for this analysis was obtained from Kaggle, specifically the Breast Cancer Wisconsin (Diagnostic) dataset (UCI Machine Learning Repository, 1995). The dataset was prepared by removing unnecessary columns and converting the target variable into a numerical format to ensure compatibility with machine learning algorithms.

The dataset consists of 569 observations and 30 numerical features describing tumor characteristics, along with one target variable. Since all features are numerical, no additional encoding was required. Summary statistics indicate variation across features, suggesting that the dataset contains meaningful patterns suitable for classification. Exploratory Data Analysis (EDA) is an essential step for understanding the structure and characteristics of the dataset before model development

4.1.1 Alignment Between Planned and Conducted Analysis

Step from Question 3 (Planning)	Description of Planned Task	What You Did in Question 4 (EDA)
Structure and characteristics	Understand dataset size, features, and data types	Used <code>df.info()</code>
Summary statistics	Analyze central tendency, spread, and variability	Used <code>df.describe()</code>
Missing values check	Ensure dataset completeness and quality	Used <code>df.isnull().sum()</code>
Class distribution	Identify balance between benign and malignant cases	Used <code>sns.countplot()</code>

Feature distribution	Examine patterns and detect skewness in variables	Plotted histograms (feature distributions)
Correlation analysis	Identify relationships and multicollinearity	Used <code>sns.heatmap()</code>
Outlier detection	Detect extreme values that may affect the model	Used <code>sns.boxplot()</code>

4.1.3 Class Distribution

The distribution of the target variable was analyzed to determine whether the dataset is balanced. The results show that the dataset contains more benign cases than malignant cases, indicating a slight class imbalance.

Analyzing class distribution is important in classification tasks, as imbalanced datasets can bias model predictions toward the majority class (Han *et al.*, 2011). The visualization was created using Python libraries such as Seaborn and Matplotlib.

4.1.4 Feature Distributions

Histograms were used to examine the distribution of selected features in the dataset. Histograms are a commonly used exploratory data analysis technique for visualizing the frequency distribution of numerical variables (IBM, 2023).

The results show that several variables are right-skewed, indicating that most values are concentrated at lower ranges with a smaller number of higher extreme values. Some features also display approximately normal distributions, while others show significant skewness. This variation suggests differences in how features are distributed across the dataset, reflecting varying statistical properties such as skewness and spread (James *et al.*, 2021).

4.1.5 Correlation Analysis

A correlation heatmap was used to analyze relationships between features. The results show strong positive correlations between variables related to tumor size, such as radius, perimeter, and area. This is visible from the high correlation values between these features, suggesting that they capture similar information.

Moderate correlations are also observed among features such as compactness, concavity, and concave points, while some variables show weaker relationships.

Identifying correlated features is important, as highly related variables may introduce multicollinearity, which can affect model performance (James *et al.*, 2021).

4.1.6 Outlier Detection

Boxplots were used to detect outliers in selected features of the dataset. The results show the presence of several extreme values, particularly in the *area_mean* feature, where multiple observations extend significantly beyond the upper whisker of the plot.

This indicates that some tumor measurements are unusually large compared to the majority of the data. In addition, *area_mean* shows a much wider range compared to features such as *radius_mean* and *texture_mean*, highlighting differences in variability across features.

These outliers suggest variability in the dataset, which may influence model performance if not carefully considered.

4.1.7 Feature Distribution by Class

To further analyze how feature distributions differ between benign and malignant cases, boxplots were used to compare *area_mean* across the two classes.

The results show a clear separation between the groups, with malignant tumors generally having higher values than benign tumors. The median and overall distribution for malignant cases are significantly higher, indicating larger tumor size.

Additionally, malignant tumors exhibit greater variability and more extreme values, while benign cases are more tightly clustered.

This clear distinction between classes suggests that *area_mean* is a strong predictor of tumor classification and contributes to the model's effectiveness.

4.2 Feature Selection

Feature selection was performed to identify the most relevant variables for predicting the diagnosis outcome. This step is important as it reduces redundancy and improves model efficiency by focusing on significant predictors (Scikit-learn, 2024; IBM, 2023).

Correlation analysis was used to measure the relationship between each feature and the target variable. Features with strong correlations, such as *concave_points_worst*, *perimeter_worst*, *radius_worst*, and *area_worst*, were identified as important predictors of cancer.

These features are primarily related to tumor size and shape, which are key indicators of malignancy. To ensure that only meaningful variables were included, a threshold of 0.5 was applied to select features with moderate to strong correlation with the target variable. This helps ensure that only variables with meaningful relationships are retained for model training, improving model performance and reducing unnecessary complexity (IBM, 2023; Scikit-learn, 2024).

Based on this analysis, features with correlation values greater than 0.5 were selected for model training, while less relevant features were excluded.

4.3 Train Model

The selected features were used to train a Random Forest classification model. The dataset was first separated into input features (X) and the target variable (y), where the selected features were used as predictors and the diagnosis variable as the outcome.

The data was then split into training and testing sets using an 80/20 split. The training dataset contained 455 observations, while the testing dataset contained 114 observations. This split ensures that the model is trained on one portion of the data and evaluated on unseen data, allowing for a fair assessment of model performance.

A Random Forest classifier was initialized and trained using the training dataset. A fixed random state was applied to ensure reproducibility of results.

Once the model was trained, it was used to generate predictions on the test dataset. The predicted values are represented as binary outputs, where 0 indicates a benign tumor and 1 indicates a malignant tumor.

For example, the model produced predictions such as:

```
[0 1 1 0 0 1 1 1 1 0]
```

5. Evaluate your model

5.1 Interpret and evaluate model

The model was evaluated using the test dataset to assess its predictive performance. As planned in Question 3.5, multiple evaluation metrics were used, including accuracy, precision, recall, F1-score, and the confusion matrix. These metrics provide a comprehensive evaluation of classification performance, particularly in medical contexts where correctly identifying cancer cases is critical (Han *et al.*, 2011).

The model achieved an accuracy of **95.6%**, indicating that it correctly classified the majority of cases. This demonstrates that the Random Forest model is highly effective in distinguishing between benign and malignant tumors.

The confusion matrix results were:

```
[[69 2]
```

```
 [ 3 40]]
```

This shows that:

- 69 benign cases were correctly classified
- 40 malignant cases were correctly classified
- Only 5 cases were misclassified

The classification report provides further details on model performance. The model achieved a precision of **0.95–0.96**, indicating that most predictions are correct. The recall for malignant cases is **0.93**, meaning that the model correctly identifies **93% of cancer cases**, which is particularly important in medical diagnosis.

The F1-score values are also high, demonstrating a good balance between precision and recall. This balance is important in medical classification problems, where both false positives and false negatives can have serious implications.

Overall, the model demonstrates strong predictive performance, with high accuracy and effective detection of malignant cases.

5.2 Retrain with different parameters (if necessary)

Although the model performs well, it is important to assess whether retraining is necessary. In this case, the model achieved a high accuracy of 95.6%, along with strong precision, recall, and F1-score values. This indicates that the model is already performing effectively in classifying both benign and malignant cases.

Given the strong performance, retraining the model is not strictly necessary, as there are very few misclassifications and the model demonstrates reliable predictive capability.

However, further improvements could still be explored through hyperparameter tuning. Adjusting parameters such as the number of trees (`n_estimators`) and the maximum depth (`max_depth`) may lead to slight improvements in performance or enhance the model's ability to generalise to new data.

Techniques such as `GridSearchCV` may be used to systematically test different parameter combinations and identify optimal settings. In addition, cross-validation could be applied to further ensure model stability and reliability.

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